



32in Gravel Bike Spring 2026 – Cal Poly Bike Builders

Designed and manufactured a custom 32-inch-wheel gravel bike frame from concept through final assembly. Developed frame geometry through reverse engineering other bikes, fabricated frame from 0.035 in chromoly tubing, and TIG welded using frame building fixtures. Designed metal 3D-printed rear dropouts integrating a SRAM UDH derailleur interface and flat-mount brake in order to reduce machining requirements while improving drivetrain and tire clearance.

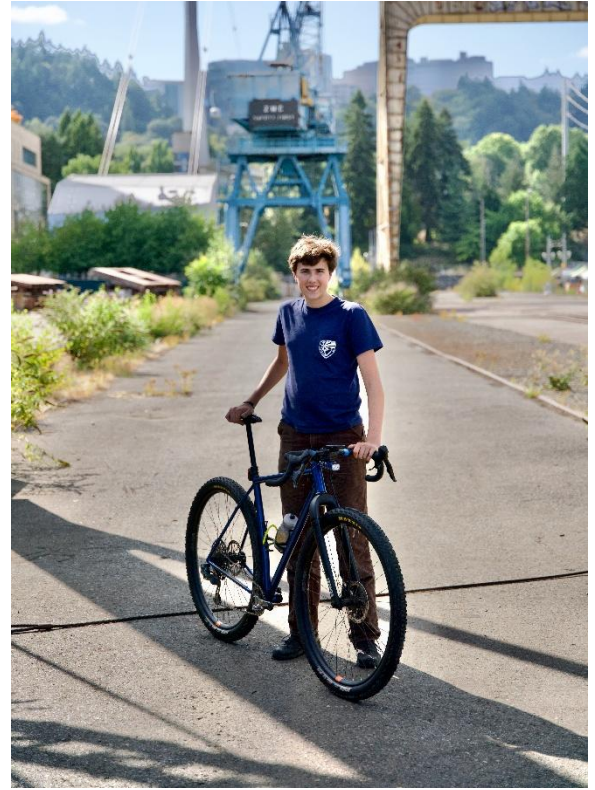
Skills: Tig welding, Metal 3d Printing, Powder Coating

Organizations: Cal Poly Bike Builders

Featured in: [Bike Rumor](#)



Photos from MADE Bike Show 2026 (Portland, OR)



MADE Bike Show 2026



Welded front triangle



Front triangle in frame jig ready for welding



Tube in milling machine for mitering



Metal 3d printed dropouts and yoke



Frame prepped for powder coating



Powdered frame



Frame after baking

FPV Flying Wing Winter 2026

Designed, fabricated, and flight-tested a 3D-printed FPV flying wing with an emphasis on durability, modularity, and serviceability. Adapted an existing airframe design for additive manufacturing to simplify replacement of structural components and electronics modules. Installed Matek F405 Wing flight controller running iNav for GPS flight modes.

Skills: 3d Printing, FPV, INAV



First iteration printed in standard PLA



Final version printed in LW-PLA being chased by an FPV Quadcopter



Historic Streetcar Electric Motor Upgrade

Summer 2026 – Lake Oswego, OR

Contributed to the modernization of a historic PCC streetcar by replacing original traction motors and resistor-based speed controls with modern brushless motors and electronics. Designed and fabricated custom motor mounts, electrical boxes and conduit. Installed the new drivetrain within the streetcar's existing mechanical constraints. Supported installation and integration of the propulsion hardware and high-voltage electrical system to enable operation without overhead power or a towed generator.

Skills: Welding, CAD

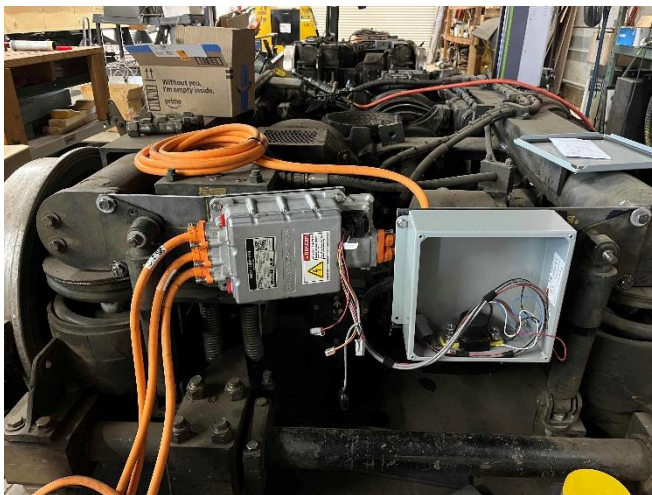
Organizations: Vintage Trolley, INC



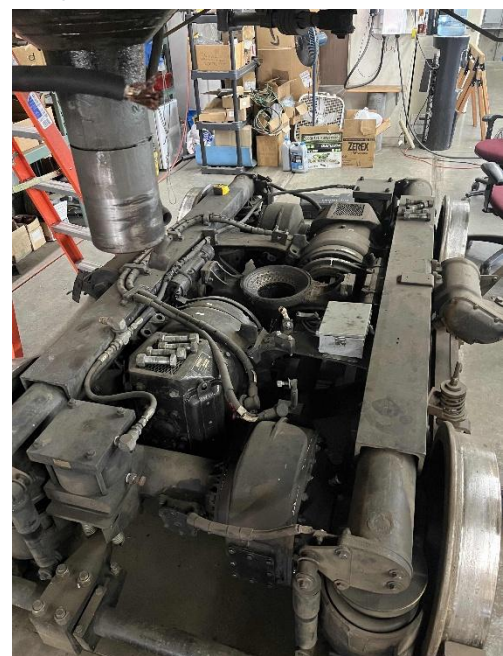
New motor mounted in PCC truck with Retention Cage



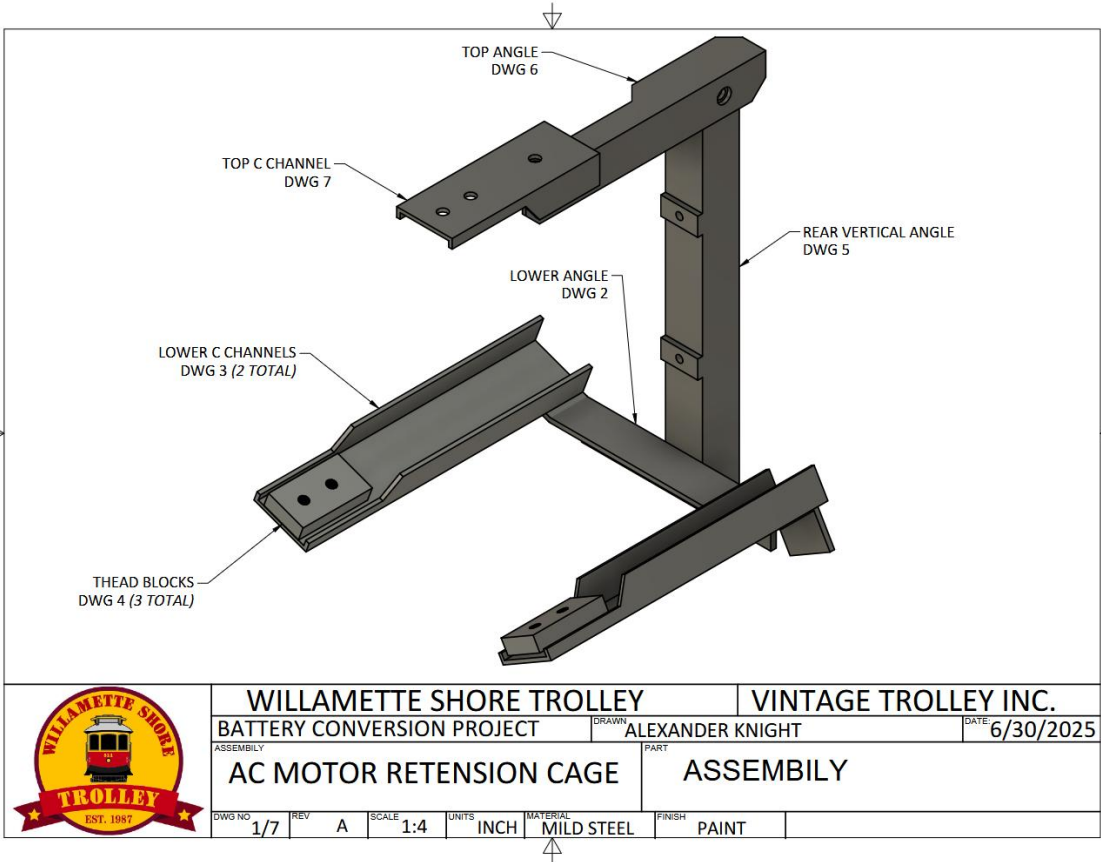
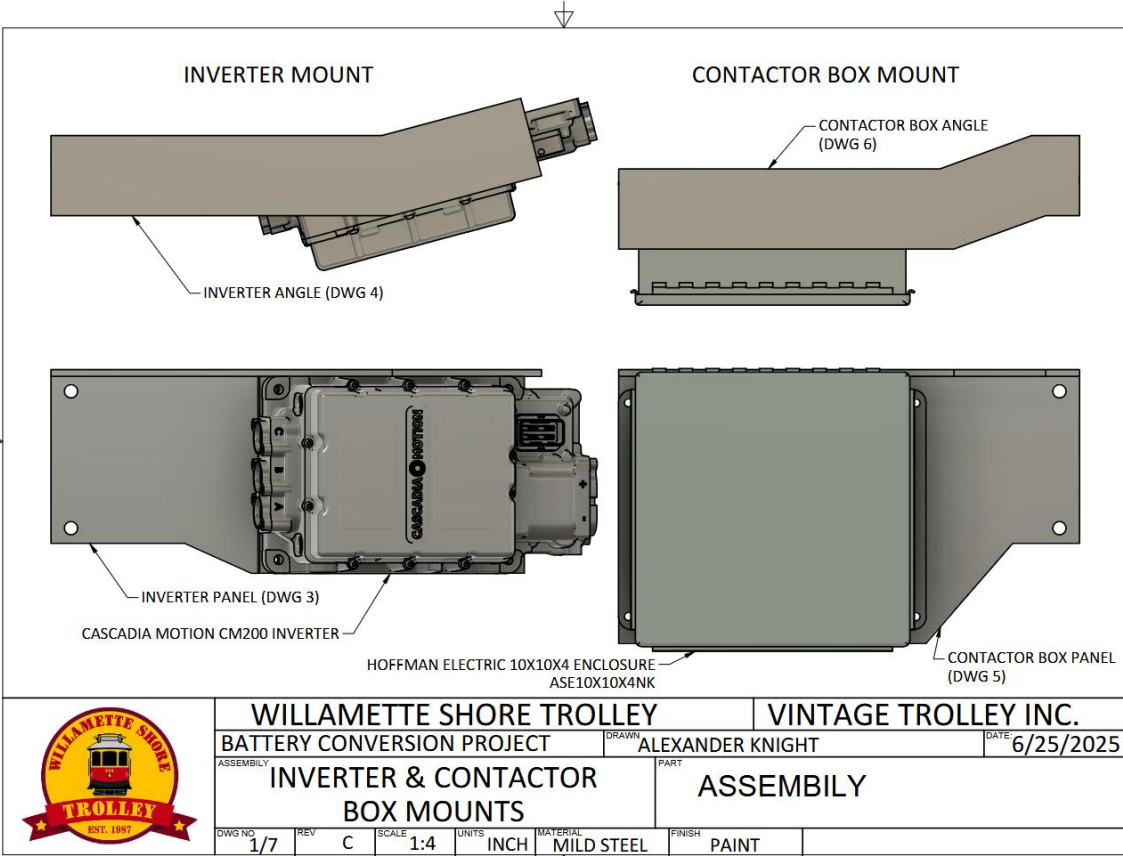
Trolley 514 (already converted to battery) next to 513 which is being converted in the Lake Oswego Trolley Barn



Motor Inverter mounted on PCC truck next to Contactor Box



Original Truck configuration with traction motors



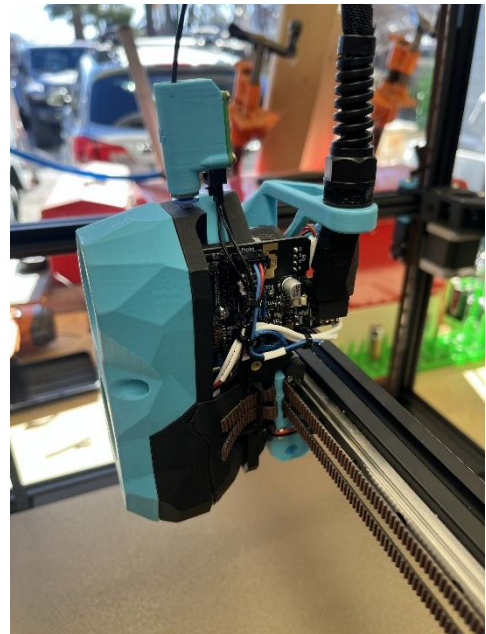
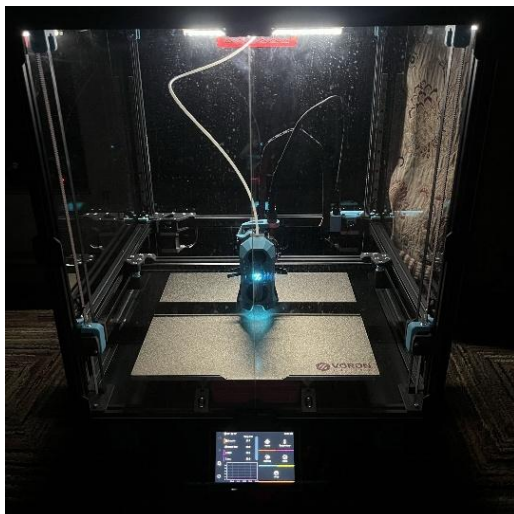
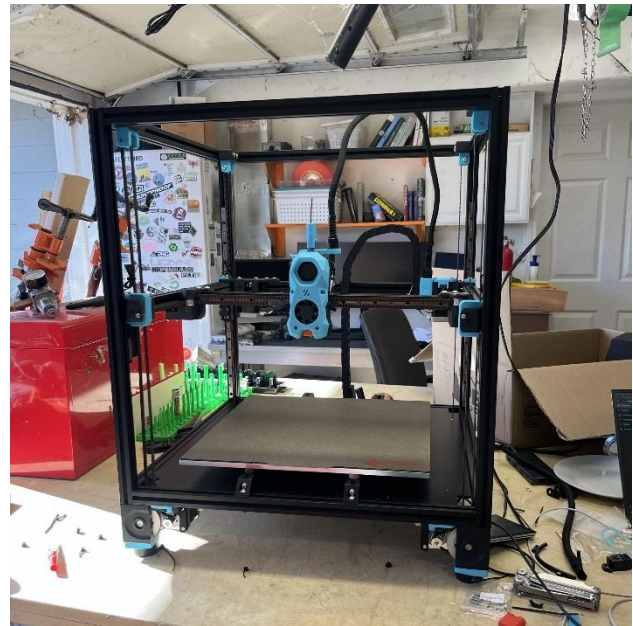
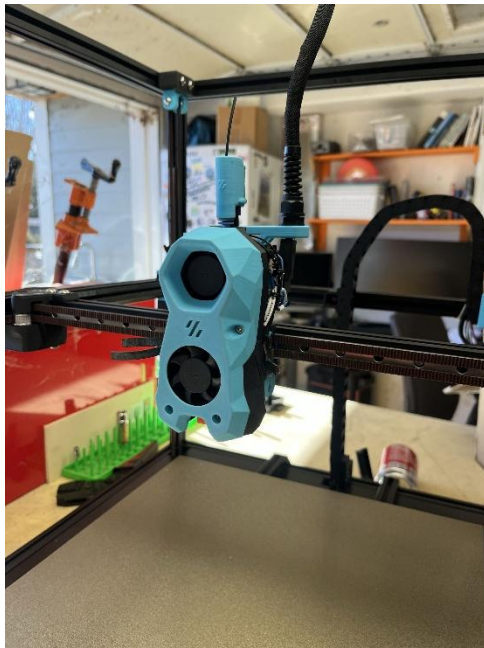


Voron 2.4 Winter 2025

Constructed, wired, configured, and calibrated a Voron 2.4 CoreXY 3D printer from individual components. Components were sourced online and printed at home and then integrated into a complete machine. Performed alignment and calibration to improve dimensional accuracy, reliability, and print quality. The project provided hands-on experience with precision machine assembly, motion control, mechatronics, and systematic troubleshooting.

Skills: 3D Printing, Linux, Soldering

Organizations: [Voron Design](https://www.vorondesign.com/)



Vannon V2 Roof Rack Winter 2024

Redesigned Vannon Roof Rack to improve manufacturability, modularity, and shipping efficiency. Reworked the existing design to make it into two parts with only the rear half being van specific. This allows shipping in shorter boxes which reduces shipping costs.

Skills: CAD, CAM, Sheet Metal Design, CNC

Organizations: [Vannon](#)



Product Photo



Support leg designed to fit a socket wrench from the outside for easy installation



Prototype test fit on a 170 Sprinter Van

Vannon V2 Running Rail Winter 2024

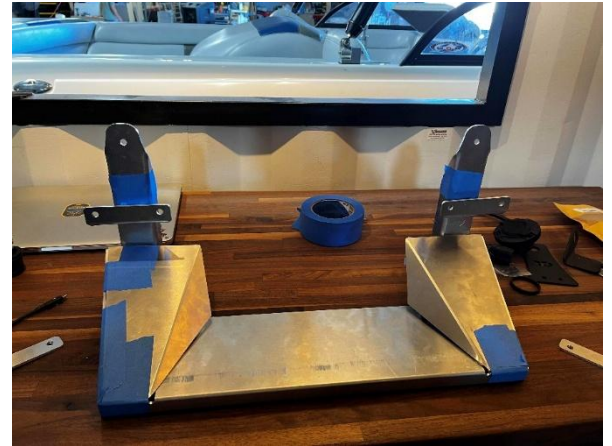
Redesigned a running rail system for Mercedes Sprinter vans to improve manufacturability, modularity, and shipping efficiency. Reworked the existing design so components could be manufactured in-house and shipped in multiple sections, reducing packaging size and overall shipping cost. The modular architecture also reduced the number of unique SKUs while allowing customers to configure the rail system to match their specific van layout and installation requirements. Included outdoor Air and Water fittings with box to hold hoses

Skills: CAD, CAM, Sheet Metal Design, CNC

Organizations: [Vannon](#)



Product Photo



First iteration of driver side step



Parts CNC cut for full van set



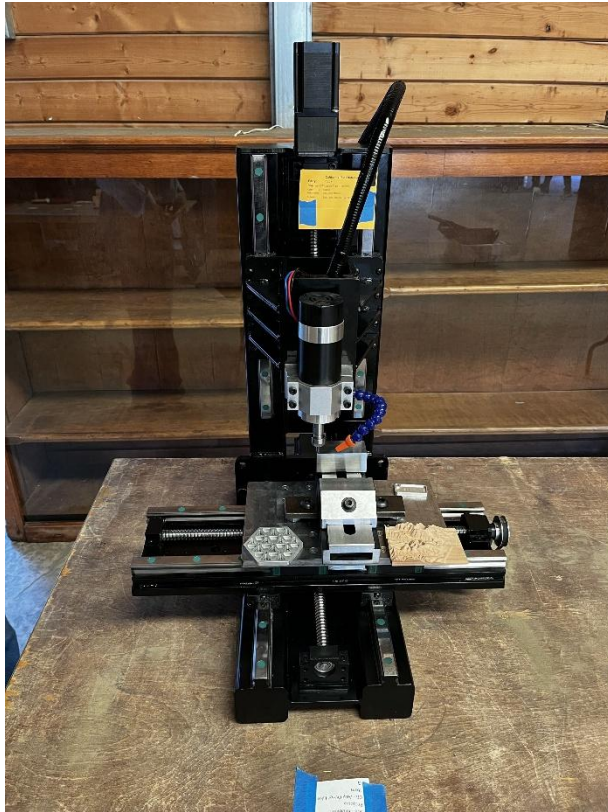
Bent Parts for passenger side

Vertical CNC Milling Machine Spring 2023

Designed and built a vertical CNC milling machine in high school metal shop. Drew the plans and built it using CNC milling, plasma cutting, a water jet. Wired with NEMA 32 stepper motors and controlled using Linux CNC on a parallel port breakout board.

Skills: CNC, CAD, CAM, Plasma Cutting, Water Jet, Welding Linux CNC, Electronics

Recognition: Best in Show at California Mid State Fair for Projects Built in a High School Shop



Finished Machine



Best in Show at California Mid State Fair



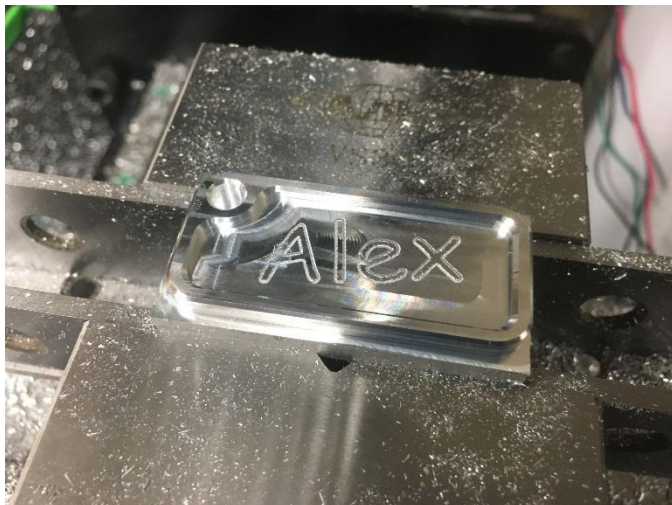
Spindle Spacer Test Fit



Calibrating Linear Rails



Test assembled before powder coat



Test part in Aluminum



Column machined parallel in Haas CNC